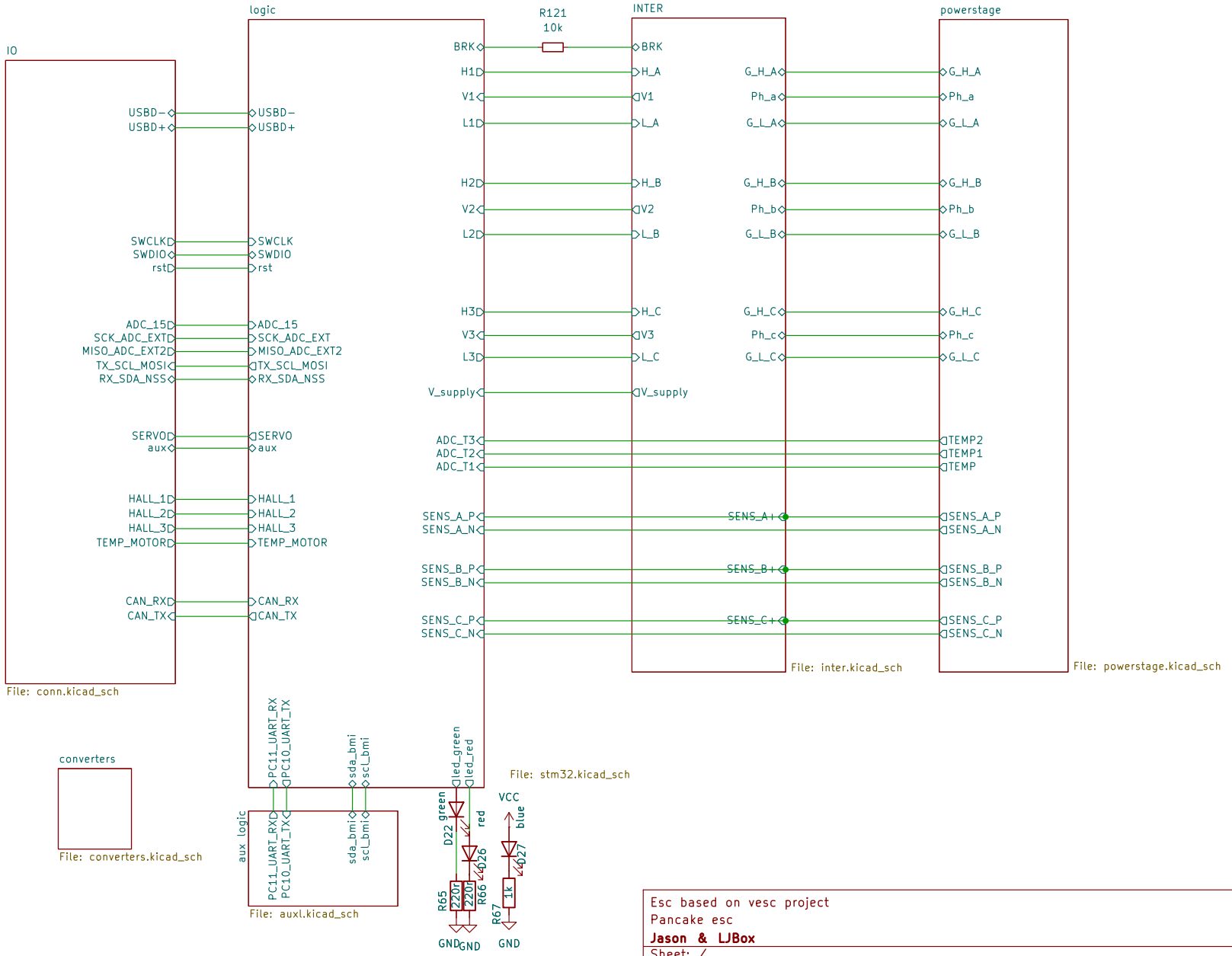
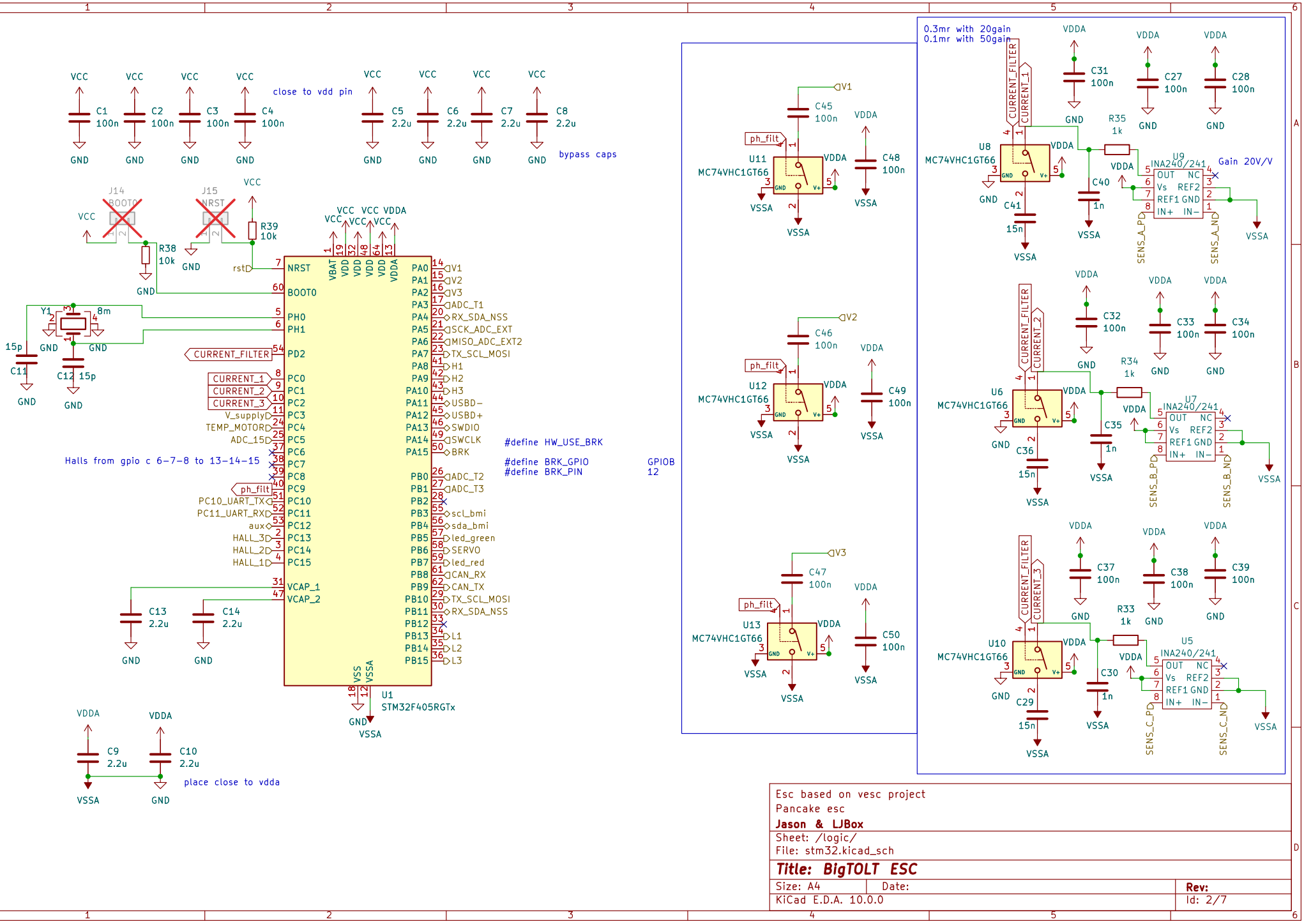


Esc based on vesc project
Pancake esc
Jason & LJBox

Sheet: /
File: tolt.kicad_sch

Title: BigTOLT ESC

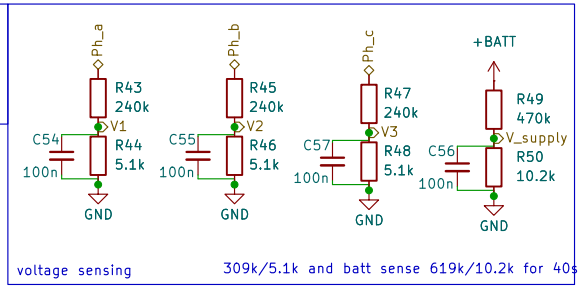
Size: A4	Date:	**Rev:**
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close to vdd pin
bypass caps
Halls from gpio c 6-7-8 to 13-14-15
#define HW_USE_BRK
#define BRK_GPIO GPIOB
#define BRK_PIN 12
place close to vdda
0.3mr with 20gain
0.1mr with 50gain
Gain 20V/V
U1
STM32F405RGTx
Esc based on vesc project
Pancake esc
Jason & LJBox
Sheet: /logic/
File: stm32.kicad_sch
Title: BigTOLT ESC
Size: A4
Date:
KiCad E.D.A. 10.0.0
Rev:
Id: 2/7

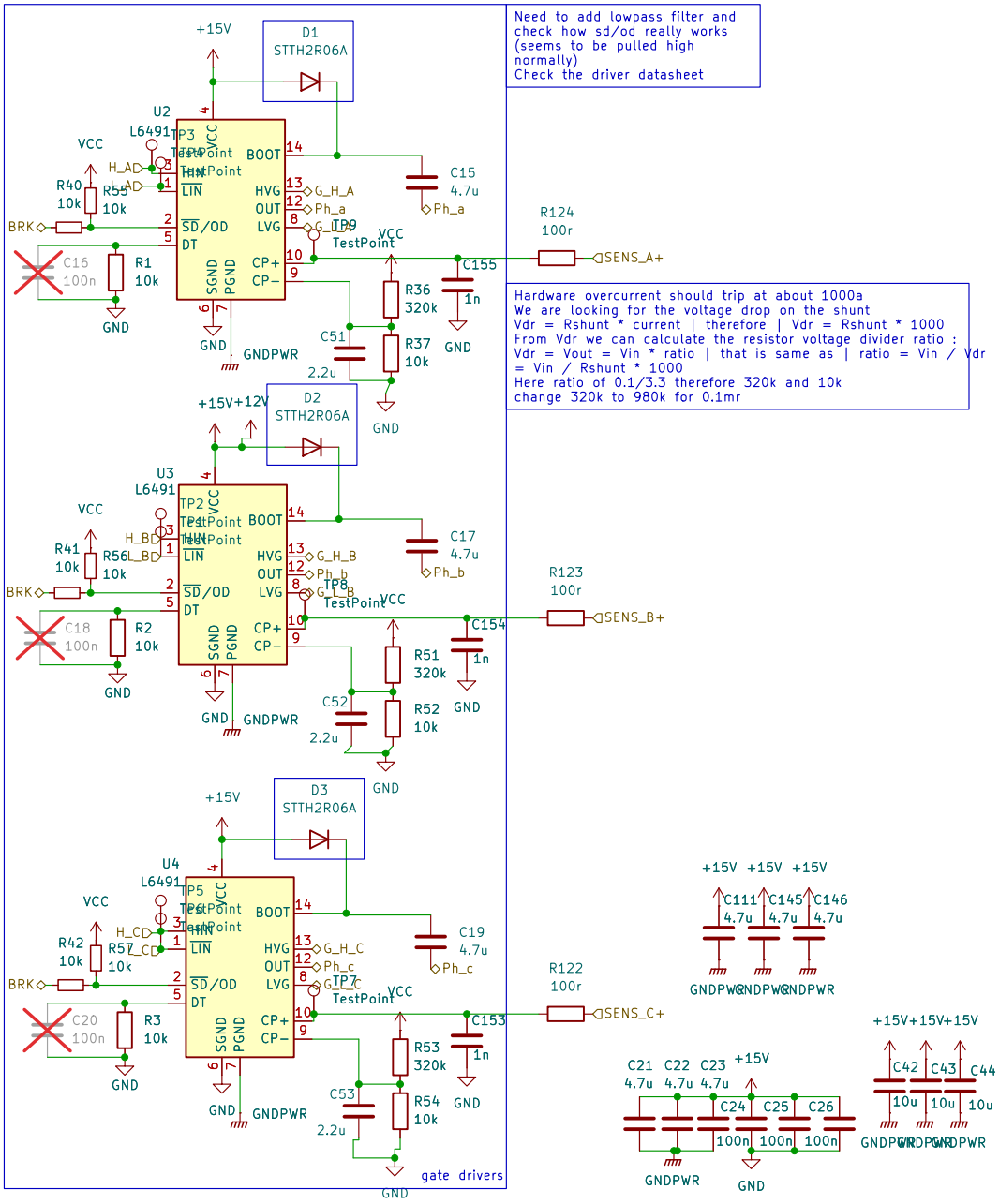
note : in hwconf it needs to have
#define INVERTED_BOTTOM_DRIVER_INPUT

Need to add lowpass filter and check how sd/od really works (seems to be pulled high normally)
Check the driver datasheet

input voltage will be up to 150v
240k and 5.1k looks good and can sense up to 158.6v



Hardware overcurrent should trip at about 1000a
We are looking for the voltage drop on the shunt
Vdr = Rshunt * current | therefore | Vdr = Rshunt * 1000
From Vdr we can calculate the resistor voltage divider ratio :
Vdr = Vout = Vin * ratio | that is same as | ratio = Vin / Vdr
= Vin / Rshunt * 1000
Here ratio of 0.1/3.3 therefore 320k and 10k
change 320k to 980k for 0.1mr



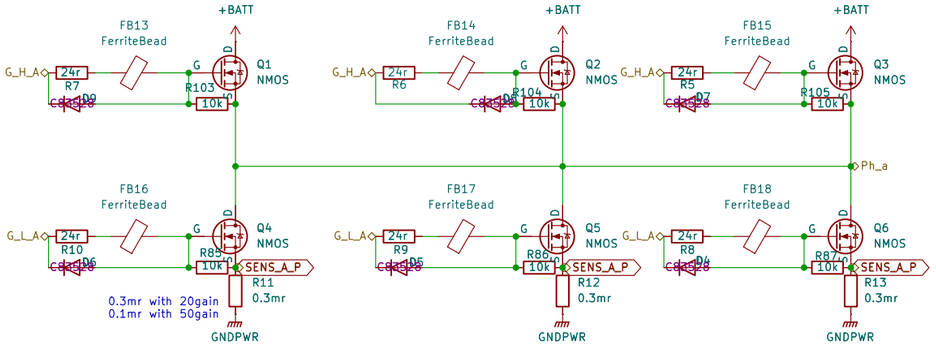
decoupling and bulk for drivers.

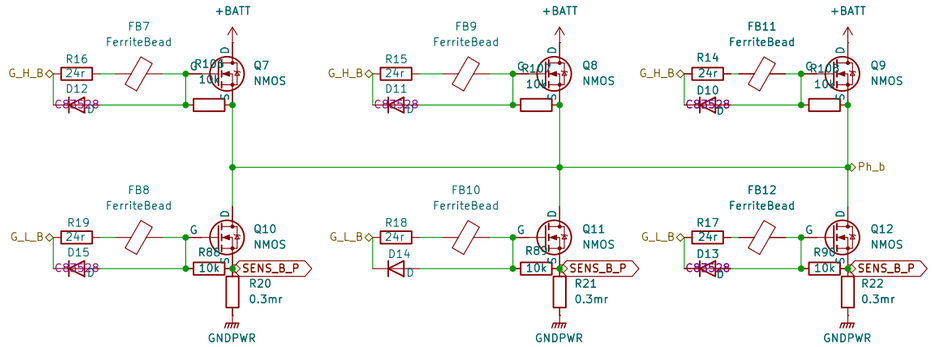
Esc based on vesc project
Pancake esc
Jason & LJBox
Sheet: /INTER/
File: inter.kicad_sch

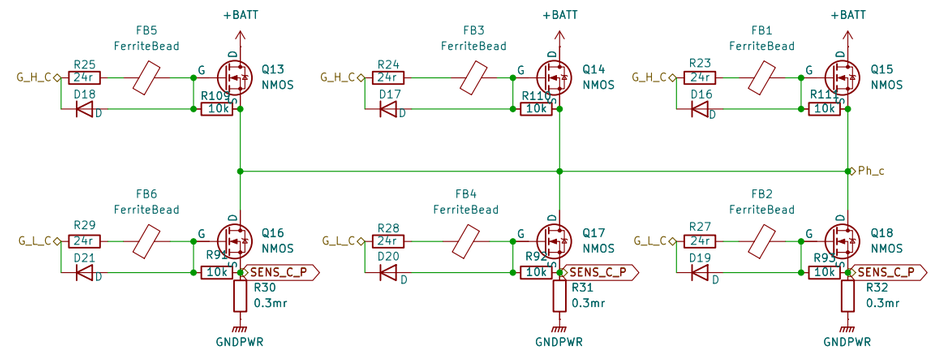
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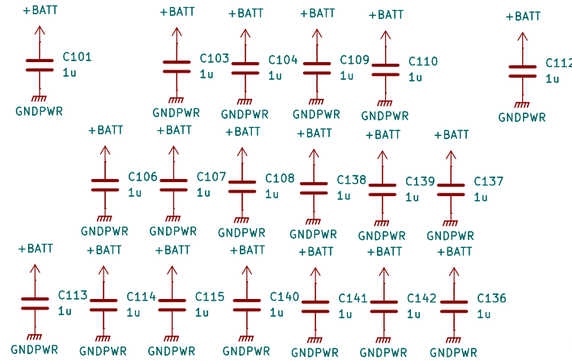
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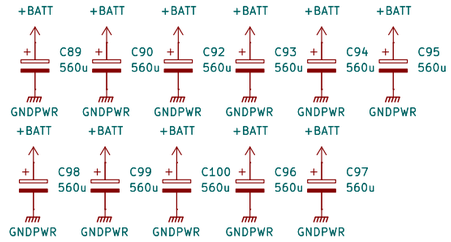
Ferrite beads value needs to be calculated according to measurements.
Note : most likely the value will be different on top and bottom gate

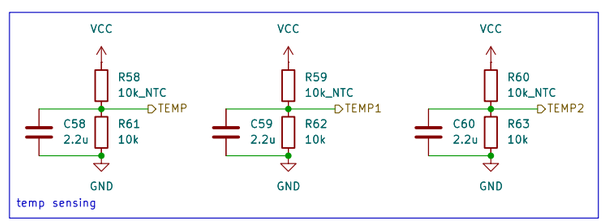




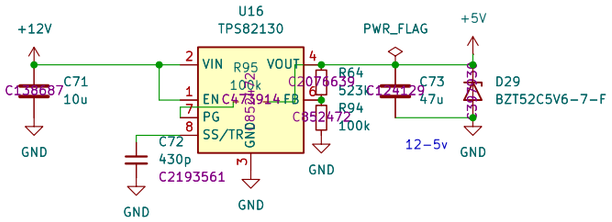


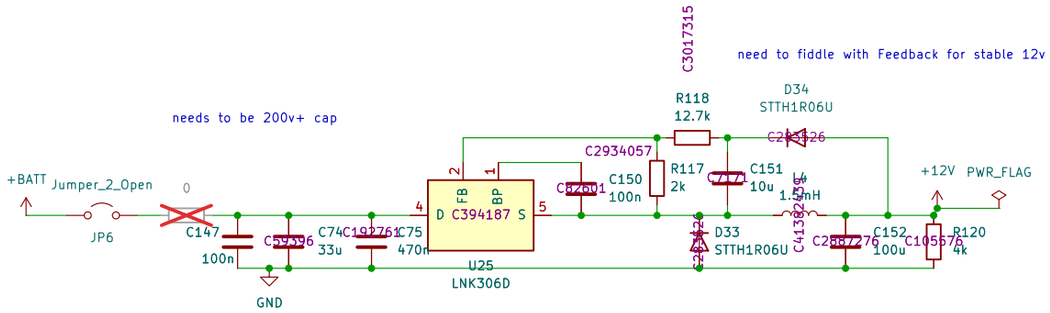


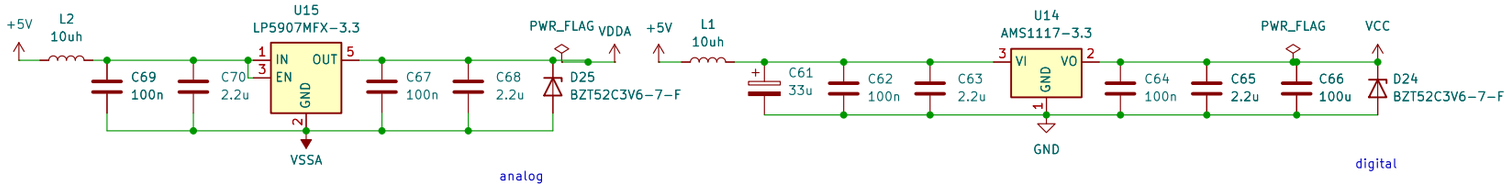


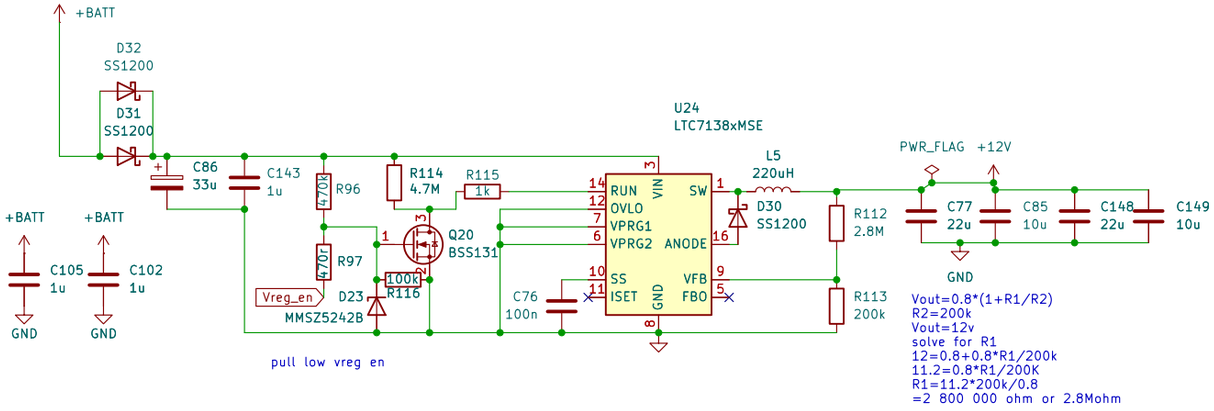


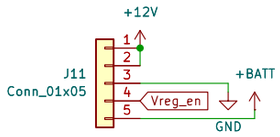
Esc based on vesc project	
Pancake esc	
Jason & LJBox	
Sheet: /powerstage/	
File: powerstage.kicad_sch	
Title: BigTOLT ESC	
Size: A3 / Date:	**Rev:**
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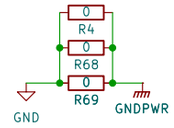










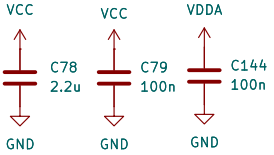


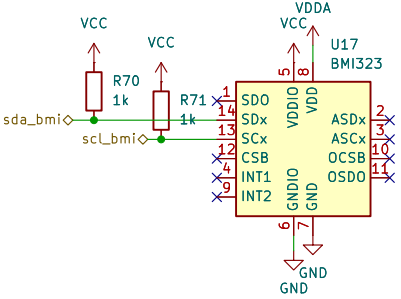
Esc based on vesc project
Pancake esc
Jason & LJBox

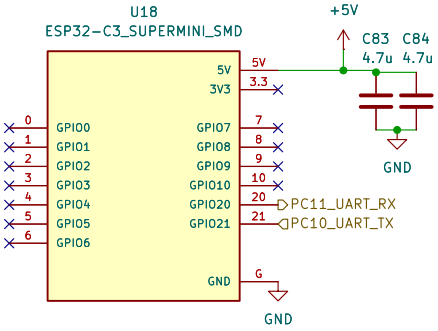
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File: converters.kicad_sch

Title: BigTOLT ESC

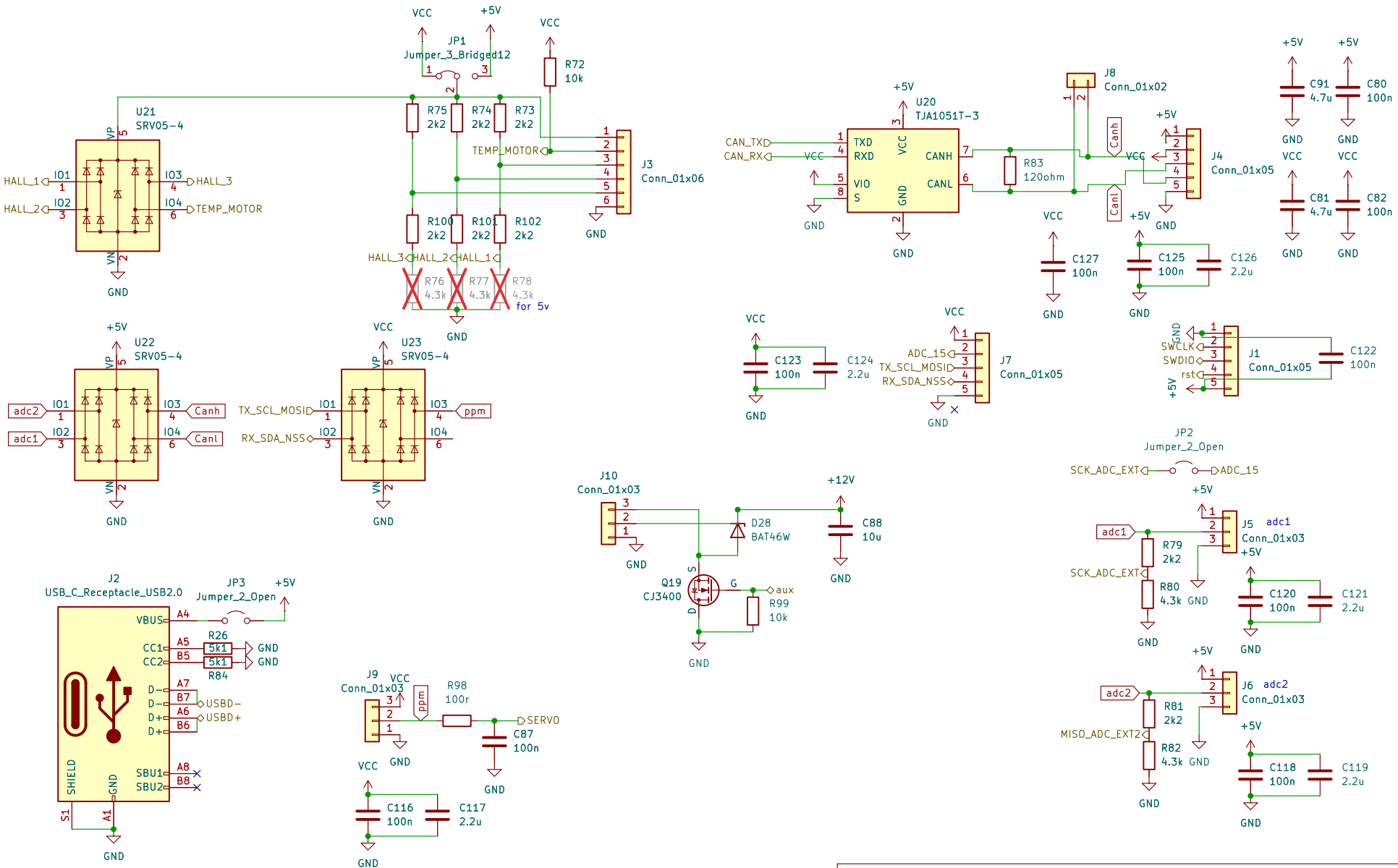
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Esc based on vesc project Pancake esc **Jason & LJBox** Sheet: /aux logic/ File: auxl.kicad_sch		
Title: BigTOLT ESC		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 6/7



Esc based on vesc project
Pancake esc
Jason & LJBox

Sheet: /IO/
File: conn.kicad_sch

Title: BigTOLT ESC

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 7/7